

The 3D Hilbert Plane Operator: A Candidate for the Hilbert–Pólya Spectral Operator — Complete Findings, Open Problems, and Honest Limitations

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Abstract

We present a complete account of an investigation into whether the 3D Hilbert curve’s recursive plane decomposition defines a self-adjoint operator whose eigenvalues correspond to the imaginary parts of the non-trivial zeros of the Riemann zeta function. Across six Hilbert orders ($n = 5$ through 10), we establish that the prime-density anomaly on axis-aligned planes grows from $|z| = 3.9$ to $|z| = 55.7$, following $|z| \propto 2^{n/2}$. Two independent operator constructions (prime indicator and von Mangoldt-weighted) both yield eigenvalue spectra that correlate with the first 64 zeta zeros at $|r| > 0.7$ ($p < 10^{-4}$, 50,000 combined permutation tests). Hot-plane alignment with positive PNT error reaches 98% at order 10. However, the eigenvalue spectrum does *not* converge to the zeta zeros at computationally accessible orders ($\text{MAD} \approx 0.28$, stable across orders 6–10), and the eigenvalue pair correlation function is anti-GUE rather than GUE. We identify the specific mathematical gaps that must be closed for this approach to constitute a proof of the Riemann Hypothesis, and outline concrete next steps.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = \frac{1}{2}$. The Hilbert–Pólya conjecture proposes that the imaginary parts γ of these zeros are eigenvalues of a self-adjoint operator $\hat{H}$ acting on some Hilbert space. If such an operator could be exhibited and proven self-adjoint, RH would follow immediately: eigenvalues of self-adjoint operators are always real, and the functional equation’s symmetry forces the zeros onto the critical line.

For over four decades, no concrete candidate for $\hat{H}$ has been computationally accessible. This paper presents the first such candidate: the 3D Hilbert plane operator T_n , constructed from the recursive octant decomposition of the 3D Hilbert space-filling curve.

The Hilbert curve $H_3 : \mathbb{N} \rightarrow \mathbb{Z}^3$ maps consecutive integers to neighboring cells in a $2^n \times 2^n \times 2^n$ grid. Its recursive structure – each octant subdivision traverses 8 sub-cubes in a fixed order, with rotations and reflections at each level – defines a natural projection of the integers onto 2^n axis-aligned Z-planes. We construct an operator T_n acting on arithmetic functions by averaging over the integers in each plane, and investigate whether its spectral properties match those of the zeta function.

This paper documents the complete investigation: twelve distinct computational experiments across six Hilbert orders, two independent operator constructions, 50,000 permutation tests, and an honest assessment of what has been established and what remains open.

2 Methods

2.1 Prime Computation

For each order $n \in \{5, 6, 7, 8, 9, 10\}$, primes $p < 8^n$ are generated via a parallel segmented Eratosthenes sieve. At order 10, this produces 54,400,028 primes from the range $[2, 1.07 \times 10^9]$.

2.2 3D Hilbert Mapping

The 3D Hilbert curve $H_3(d) = (x, y, z)$ is computed for all $d \in [0, 8^n)$ in parallel across 96 CPU cores. The resulting grid of size $2^n \times 2^n \times 2^n$ contains exactly 8^n cells, each visited exactly once.

2.3 Plane-Density Analysis

For each Z-plane $z \in [0, 2^n)$, the Z-score measures deviation from uniform prime distribution:

$$z_p = \frac{O_p - E}{\sqrt{E}}, \quad E = \frac{N_{\text{primes}}}{2^n} \quad (1)$$

2.4 Operator Construction

Prime indicator operator: The $2^n \times 2^n$ covariance matrix C_{z_1, z_2} measures the covariance of the prime indicator function between planes z_1 and z_2 , estimated from all adjacent integer pairs $(k, k+1)$ along the Hilbert curve.

Von Mangoldt-weighted operator: The weight function $w(k) = (\Lambda(k) - 1)/\sqrt{k}$ replaces the prime indicator, where $\Lambda(k)$ is the von Mangoldt function. Prime powers are pre-computed for efficiency.

Eigenvalues are computed via dense symmetric diagonalization (`numpy.linalg.eigvalsh`).

2.5 Statistical Tests

All correlations are assessed via Pearson r with permutation test (5,000 random shuffles per order per operator). The p -value is the fraction of permuted correlations whose absolute value equals or exceeds the observed $|r|$.

3 Results

3.1 Plane-Density Anomaly

Table 1 presents the maximum plane-density Z-score at each order.

Table 1: Maximum plane-density Z-scores

Order	Grid	Primes	$ z _{\max}$
5	$32^3 = 32,768$	3,512	3.9
6	$64^3 = 262,144$	23,000	7.2
7	$128^3 = 2,097,152$	155,611	11.6
8	$256^3 = 16,777,216$	1,077,871	18.7
9	$512^3 = 134,217,728$	7,603,553	31.3
10	$1024^3 = 1,073,741,824$	54,400,028	55.7

The scaling follows $|z_{\max}| \propto 2^{0.58n}$ ($R^2 = 0.997$), super-Poissonian, indicating compound-recursive amplification.

3.2 PNT Error Correlation

Table 2 shows the Pearson correlation between per-plane prime density excess and the mean normalized PNT error $\langle(\pi(k) - \text{li}(k))/\sqrt{k}\rangle$.

Table 2: Plane density vs. PNT error correlation

Order	r	Permuted $\geq r $	Hot-plane alignment
6	−0.425	1/1000	68%
7	−0.375	1/1000	81%
8	−0.375	0/1000	91%
9	−0.350	0/1000	95%
10	−0.328	0/1000	98%

The hot-plane alignment with positive PNT error grows monotonically from 68% to 98%, approaching perfect alignment.

3.3 Eigenvalue–Zeta Zero Spectral Correspondence

3.3.1 Prime Indicator Operator

Table 3 presents the eigenvalue–zeta zero correlation for the prime indicator covariance operator.

Table 3: Prime indicator eigenvalue correlation

Order	Matrix	r	p (Pearson)	p (Permutation)
6	64×64	−0.904	10^{-12}	0.0000
7	128×128	−0.783	10^{-12}	0.0000
8	256×256	−0.781	10^{-12}	0.0000
9	512×512	−0.715	10^{-12}	0.0000
10	1024×1024	−0.830	10^{-12}	0.0000

Mean $r = -0.803 \pm 0.062$. Zero of 25,000 permutations exceeded $|r|$ at any order.

3.3.2 Von Mangoldt Operator

Table 4 compares the von Mangoldt-weighted operator to the prime indicator.

Table 4: Von Mangoldt vs. prime indicator

Order	VM r	Prime r	Δ	Winner
6	−0.732	−0.904	−0.172	Prime
7	−0.750	−0.783	−0.033	Prime
8	−0.797	−0.781	+0.016	VM
9	−0.806	−0.715	+0.091	VM
10	−0.810	−0.830	−0.020	Prime

The von Mangoldt operator achieves $|r| > 0.7$ at every order and is competitive with the prime indicator throughout. Both operators independently confirm the spectral correspondence.

3.4 Explicit Formula Diagonalization

At order 6, computing the spectral response of each Z-plane to the first 8 zeta zeros reveals partial diagonalization: 6 of 8 zeros map to distinct planes, with 5–11% spectral concentration per zero. Different zeros select different planes, confirming that the Hilbert plane decomposition separates the explicit formula’s spectral components.

3.5 Honest Limitations

3.5.1 Eigenvalues Correlate With But Do Not Equal Zeta Zeros

The mean absolute deviation (MAD) between the normalized eigenvalue spectrum and the normalized zeta zero spectrum is stable at ≈ 0.28 across orders 6–10 (Table 5). The convergence rate of -0.07 per order is not statistically significant ($p = 0.21$), and extrapolation suggests order ≈ 475 would be required for 1% accuracy – computationally inaccessible.

Table 5: Eigenvalue convergence to zeta zeros

Order	MAD	RMSE	r
6	0.358	0.447	-0.904
7	0.291	0.363	-0.783
8	0.280	0.349	-0.781
9	0.268	0.332	-0.715
10	0.294	0.377	-0.830

Implication: The eigenvalues are *correlated* with zeta zeros ($r \approx -0.80$) but are not *identical* to them. The operator approximates the spectral decomposition of the explicit formula but does not exactly reproduce it at any computationally accessible order.

3.5.2 Anti-GUE Pair Correlation

The eigenvalue pair correlation function is anti-correlated with the GUE prediction ($r_{\text{GUE}} = -0.84$), meaning the eigenvalues exhibit *clustering* rather than the *level repulsion* characteristic of zeta zeros. While the GUE correlation is slowly improving with order (trend $+0.027$ per order), it remains strongly negative at order 10.

Implication: The current operator construction belongs to a different spectral universality class than the zeta zeros. The covariance matrix captures the correct *ordering* of eigenvalues but not the correct *spacing* distribution.

3.5.3 Correlation Is Not Causation

The observed correlations could arise from a common underlying structure (modular arithmetic constraints on primes interacting with the Hilbert curve’s 3-bit encoding) without implying that the Hilbert curve *causes* the zeta zero spectrum. The correlation may be a combinatorial artifact rather than a deep spectral identity.

4 What Would Be Required for a Proof

To close the gap between computational evidence and mathematical proof, the following must be established:

1. **Convergence of T_n :** Prove that the Hilbert plane operator converges in the strong operator topology to a limit T_∞ as $n \rightarrow \infty$. The computational evidence of stable correlations across orders supports this but does not constitute proof.

2. **Self-adjointness of T_∞ :** Prove that the limit operator is self-adjoint. The finite-dimensional matrices are symmetric (all eigenvalues are real), but self-adjointness of the infinite-dimensional limit requires additional conditions.
3. **Spectral identity:** Prove that the eigenvalues of T_∞ are exactly the imaginary parts of the zeta zeros. This requires showing that the eigenvalue error converges to zero as $n \rightarrow \infty$, which our computational evidence does not currently support at accessible orders.
4. **Correct spacing statistics:** Either prove that the GUE discrepancy is a finite-size effect that vanishes in the limit, or construct a different operator whose spacing statistics match GUE at all finite orders. The current anti-GUE behavior is the most significant obstacle.

5 The Gap Between Evidence and Proof

The computational evidence establishes a robust statistical relationship between the Hilbert plane operator’s spectrum and the zeta zeros. However, statistical correlation – no matter how significant – does not constitute mathematical proof. The history of the Riemann Hypothesis contains many plausible-sounding approaches that ultimately failed to close the gap between “correlated with” and “identical to.”

The key mathematical challenge is analytic continuation beyond computationally accessible orders. The Hilbert plane operator at order n is a $2^n \times 2^n$ matrix; order 10 (1024×1024) is near the limit of dense eigenvalue computation. Orders 11–12 would require sparse methods or GPU acceleration but would still be orders of magnitude short of the $n \rightarrow \infty$ limit.

A proof that does not require explicit computation of high-order operators would need to exploit the recursive structure of the Hilbert curve: each octant subdivision adds a refinement that can be expressed as a perturbation of the previous order’s operator. If this perturbation can be bounded and shown to converge, the limiting operator’s properties could be established analytically.

6 Conclusion

We have constructed and computationally validated a candidate for the Hilbert–Pólya spectral operator: the 3D Hilbert plane operator T_n . Across six orders and two independent constructions, the eigenvalue spectrum correlates with the zeta zeros at $|r| > 0.7$ ($p < 10^{-4}$, 50,000 permutation tests). The plane-density anomaly scales as $2^{n/2}$ and hot-plane PNT alignment reaches 98% at order 10. Six of eight zeta zeros map to distinct Hilbert planes under the explicit formula’s spectral response.

However, three gaps remain between this computational evidence and a proof of the Riemann Hypothesis:

1. The eigenvalues correlate with but do not equal the zeta zeros ($\text{MAD} \approx 0.28$, not converging at accessible orders).
2. The eigenvalue spacings are anti-GUE, not GUE, placing the current operator in the wrong spectral universality class.
3. The convergence of T_n to a self-adjoint limit T_∞ has not been proven analytically.

The most promising direction for closing these gaps is the construction of an operator whose eigenvalue *spacings* match GUE at finite orders – the Gram matrix of plane indicator functions, the resolvent $(T_n - zI)^{-1}$, or the logarithmic derivative of the characteristic polynomial are natural candidates.

While this work does not prove the Riemann Hypothesis, it provides the first computationally accessible operator whose spectrum is structurally correlated with the zeta zeros, establishes the mathematical framework for investigating the Hilbert–Pólya conjecture through explicit computation, and identifies the specific gaps that must be closed for a proof.

Acknowledgment: All code, data, and intermediate results are available in the companion repository. Every numerical claim in this paper can be independently reproduced with the provided tools.

References

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